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Camera optical lens

Abstract

A camera optical lens is provided, including from an object side to an image side: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; and a seventh lens, the camera optical lens satisfies following conditions: $1.30 \leq f_7/f \leq 4.00$; $1.70 \leq n_1$; $2.00 \leq f_{34}/f \leq 7.00$; and $120.00 \leq (FOV \times f)/IH$, where f denotes a focal length of the camera optical lens; f_7 denotes a focal length of the seventh lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; n_1 denotes a refractive index of the first lens; FOV denotes a field of view of the camera optical lens; and IH denotes an image height of the camera optical lens. The camera optical lens has good optical performance such as large aperture, ultra-thinness and wide angle while ensuring miniaturization, small chromatic aberration, small lateral color, and high resolution up to 5M.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of optical lens and, in particular, to a camera optical lens suitable for handheld terminal devices such as smart phones or digital cameras, and imaging devices such as monitors or PC lenses.

BACKGROUND

(2) With the rise of smart phones in recent years, the demand for miniature camera lens is continuously increasing, and since the pixel size of the photo sensors become smaller, along with the current development trend of electronic products that requires better performance such—as thinner and smaller dimensions, miniature camera lenses with good imaging quality therefore have

become a mainstream in the market. In order to obtain better imaging quality, a structure with multiple lenses is generally used. Moreover, with development of technology and increase in diversified requirements of users, a camera lens constituted by seven lenses gradually appears in camera lens design, in case that pixel area of the photosensitive device is continuously reduced and requirements on imaging quality is continuously increased. A wide-angle camera optical lens having good optical performance is urgently needed.

SUMMARY

(3) In view of the above problems, the present disclosure provides a camera optical lens, which has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

(4) Embodiments of the present disclosure provide a camera optical lens, including from an object side to an image side: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; and a seventh lens, the camera optical lens satisfies following conditions: $1.30 \leq f_7/f \leq 4.00$; $1.70 \leq n_1$; $2.00 \leq f_{34}/f \leq 7.00$; and $120.00 \leq (FOV \times f)/IH$, where f denotes a focal length of the camera optical lens; f_7 denotes a focal length of the seventh lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; n_1 denotes a refractive index of the first lens; FOV denotes a field of view of the camera optical lens; and IH denotes an image height of the camera optical lens.

(5) As an improvement, the camera optical lens satisfies a following condition: $1.00 \leq f_5/f \leq 3.00$, where f_5 denotes a focal length of the fifth lens.

(6) As an improvement, the camera optical lens satisfies following conditions: $d_{10}=0$; and $0.40 \leq |f_5/f_6| \leq 3.00$, where d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens; f_5 denotes a focal length of the fifth lens; and f_6 denotes a focal length of the sixth lens.

(7) As an improvement, the camera optical lens satisfies a following condition: $0.35 \leq BF/TL$, where BF denotes an on-axis distance from an image-side surface of the seventh lens to an image plane of the camera optical lens; and TL denotes an on-axis distance from an object-side surface of the first lens to the image-side surface of the seventh lens.

(8) As an improvement, the camera optical lens satisfies a following condition: $0.01 \leq (d_4+d_0)/TTL \leq 0.10$, where d_4 denotes an on-axis distance from an image-side surface of the second lens to an aperture; d_0 denotes an on-axis distance from the aperture to an object-side surface of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(9) As an improvement, the camera optical lens satisfies a following condition: $0 < D/IH/FOV \leq 0.10$, where D denotes a light passing diameter of an object-side surface of the first lens corresponding to a maximum field of view of the camera optical lens.

(10) As an improvement, the camera optical lens satisfies a following condition: $0 < TTL/IH/FOV \leq 0.06$, where TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(11) As an improvement, the camera optical lens satisfies following conditions: $d_6=0$; and $d_{10}=0$, where d_6 denotes an on-axis distance from an image-side surface of the third lens to an object-side surface of the fourth lens; and d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens.

(12) As an improvement, an object-side surface of the first lens is convex in a paraxial region, and an image-side surface of the first lens is concave in the paraxial region; the first lens has a negative refractive power; and the camera optical lens satisfies following conditions: $-3.09 \leq f_1/f \leq -0.60$; $0.57 \leq (R_1+R_2)/(R_1-R_2) \leq 1.88$; and $0.02 \leq d_1/TTL \leq 0.05$, where f_1 denotes a focal length of the first lens; R_1 denotes a central curvature radius of the object-side surface of the first lens; R_2 denotes a central curvature radius of the image-side surface of the first lens; d_1 denotes an on-axis thickness of the first lens; and TTL denotes a total optical length from an object-side surface of the first lens

to an image plane of the camera optical lens along an optic axis.

(13) As an improvement, the camera optical lens satisfies following conditions: $-10.46 \leq f_2/f \leq 19.43$; $-4.44 \leq (R_3+R_4)/(R_3-R_4) \leq 22.86$; and $0.03 \leq d_3/TTL \leq 0.09$, where f_2 denotes a focal length of the second lens; R_3 denotes a central curvature radius of an object-side surface of the second lens; R_4 denotes a central curvature radius of an image-side surface of the second lens; d_3 denotes an on-axis thickness of the second lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(14) As an improvement, the camera optical lens satisfies following conditions: $-9.86 \leq f_3/f \leq 1.77$; $-1.65 \leq (R_5+R_6)/(R_5-R_6) \leq 0.53$; and $0.01 \leq d_5/TTL \leq 0.14$, where f_3 denotes a focal length of the third lens; R_5 denotes a central curvature radius of an object-side surface of the third lens; R_6 denotes a central curvature radius of an image-side surface of the third lens; d_5 denotes an on-axis thickness of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(15) As an improvement, the camera optical lens satisfies following conditions: $-3.27 \leq f_4/f \leq 2.38$; $-0.14 \leq (R_7+R_8)/(R_7-R_8) \leq 1.27$; and $0.01 \leq d_7/TTL \leq 0.23$, where f_4 denotes a focal length of the fourth lens; R_7 denotes a central curvature radius of an object-side surface of the fourth lens; R_8 denotes a central curvature radius of an image-side surface of the fourth lens; d_7 denotes an on-axis thickness of the fourth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(16) As an improvement, an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of the fifth lens is convex in the paraxial region; the fifth lens has positive refractive power; the camera optical lens satisfies following conditions:

$-0.42 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 0.84$; and $0.05 \leq d_9/TTL \leq 0.22$, where R_9 denotes a central curvature radius of an object-side surface of the fifth lens; R_{10} denotes a central curvature radius of an image-side surface of the fifth lens; d_9 denotes an on-axis thickness of the fifth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(17) As an improvement, an object-side surface of the sixth lens is concave in a paraxial region; the sixth lens has negative refractive power; the camera optical lens satisfies following conditions:

$-2.88 \leq f_6/f \leq -0.76$; $-2.78 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 0.42$; and $0.01 \leq d_{11}/TTL \leq 0.16$, where f_6 denotes a focal length of the sixth lens; R_{11} denotes a central curvature radius of an object-side surface of the sixth lens; R_{12} denotes a central curvature radius of an image-side surface of the sixth lens; d_{11} denotes an on-axis thickness of the sixth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(18) As an improvement, an object-side surface of the seventh lens is convex in a paraxial region and an image-side surface of the seventh lens is convex in the paraxial region; the seventh lens has positive refractive power; the camera optical lens satisfies following conditions:

$-0.50 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 0.28$; and $0.05 \leq d_{13}/TTL \leq 0.20$, where R_{13} denotes a central curvature radius of an object-side surface of the seventh lens; R_{14} denotes a central curvature radius of an image-side surface of the seventh lens; d_{13} denotes an on-axis thickness of the seventh lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

(19) As an improvement, at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, and the seventh lens is made of glass material.

(20) The present disclosure has following beneficial effects: the camera optical lens of the present disclosure has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost, which is especially suitable for mobile phone camera lens components and WEB camera lens that are composed of high-pixel CCD, CMOS and other imaging elements, respectively.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Many aspects of the exemplary embodiments may be better understood with reference to following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.
- (2) FIG. 1 is a structural schematic diagram of a camera optical lens according to Embodiment 1 of the present disclosure;
- (3) FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1;
- (4) FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1;
- (5) FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1;
- (6) FIG. 5 is a structural schematic diagram of a camera optical lens according to Embodiment 2 of the present disclosure;
- (7) FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5;
- (8) FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5;
- (9) FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5;
- (10) FIG. 9 is a structural schematic diagram of a camera optical lens according to Embodiment 3 of the present disclosure;
- (11) FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9;
- (12) FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9;
- (13) FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9;
- (14) FIG. 13 is a structural schematic diagram of a camera optical lens according to Embodiment 4 of the present disclosure;
- (15) FIG. 14 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 13;
- (16) FIG. 15 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 13;
- (17) FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 13;
- (18) FIG. 17 is a structural schematic diagram of a camera optical lens according to Embodiment 5 of the present disclosure;
- (19) FIG. 18 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 17;
- (20) FIG. 19 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 17;
- (21) FIG. 20 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 17;
- (22) FIG. 21 is a structural schematic diagram of a camera optical lens according to Embodiment 6 of the present disclosure;
- (23) FIG. 22 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 21;
- (24) FIG. 23 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 21;
- (25) FIG. 24 is a schematic diagram of a field curvature and a distortion of the camera optical lens

shown in FIG. 21;

(26) FIG. 25 is a structural schematic diagram of a camera optical lens according to Comparative Example 1 of the present disclosure;

(27) FIG. 26 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 25;

(28) FIG. 27 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 25; and

(29) FIG. 28 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 25.

DESCRIPTION OF EMBODIMENTS

(30) In order to better illustrate the objectives, technical solutions and advantages of the present disclosure, the present disclosure will be described in further detail below with reference to the accompanying drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the present disclosure but are not used to limit the present disclosure.

Embodiment 1

(31) Referring to FIG. 1, the present disclosure provides a camera optical lens 10. FIG. 1 is a structural schematic diagram of a camera optical lens 10 according to Embodiment 1 of the present disclosure. The camera optical lens 10 includes seven lenses. The camera optical lens 10 includes, from an object side at the left to an image side at the right, a first lens L1, a second lens L2, an aperture S1, a third lens L3, a fourth lens L4, a fifth lens L5, a sixth lens L6, and a seventh lens L7. An optical element such as an optical filter GF may be arranged between the seventh lens L7 and an image plane Si.

(32) In this embodiment, the first lens L1, the second lens L2, the third lens L3, the fourth lens L4, the fifth lens L5, the sixth lens L6, and the seventh lens L7 are each made of a glass material. In other embodiments, the lenses may also be made of other materials.

(33) A focal length of the camera optical lens 10 is defined as f , and a focal length of the seventh lens L7 is defined as f_7 . A following condition is satisfied: $1.30 \leq f_7/f \leq 4.00$. This condition specifies a ratio of the focal length f_7 of the seventh lens L7 to the focal length f of the camera optical lens 10. Within the range of the above condition, it is beneficial to receive light and ensure the amount of light passing therethrough, so that the camera optical lens 10 has a small CRA (an incidence angle of main light toward the image plane is defined as CRA), and $CRA < 17.5^\circ$.

(34) A refractive index of the first lens L1 is defined as n_1 . A following condition is satisfied: $1.70 \leq n_1$. This condition specifies a range of refractive index of the material used in the first lens L1. Within the range of the above condition, it is beneficial to reducing a diameter of front opening and improving imaging quality.

(35) The focal length of the camera optical lens 10 is defined as f , and a combined focal length of the third lens L3 and the fourth lens L4 is defined as f_{34} . A following condition is satisfied: $2.00 \leq f_{34}/f \leq 7.00$. This condition specifies a ratio of the combined focal length f_{34} of the third lens L3 and the fourth lens L4 to the focal length f of the camera optical lens 10. Within the range of the above condition, the orientation of light between the first lens L1 and the fourth lens L4 can be effectively controlled, and the aberration caused by the large-angle light passing through the first lens L1 can be reduced, while the camera optical lens 10 has a compact structure, which is beneficial to miniaturization.

(36) The focal length of the camera optical lens 10 is defined as f , a field of view of the camera optical lens 10 is defined as FOV, an image height of the camera optical lens 10 is defined as IH. A following condition is satisfied: $120.00 \leq (FOV \times f)/IH$. This condition specifies a ratio of product of the field of view FOV of the camera optical lens 10 and the focal length f of the camera optical lens 10 to the image height IH of the camera optical lens 10. Within the range of the above condition, the camera optical lens 10 can be designed with both long focal length and large field of view.

- (37) The focal length of the camera optical lens **10** is defined as f , and the focal length of the fifth lens **L5** is defined as f_5 . A following condition is satisfied: $1.00 \leq f_5/f \leq 3.00$. This condition specifies a ratio of the focal length f_5 of the fifth lens **L5** to the focal length f of the camera optical lens **10**. By controlling the focal length reasonably, within the range of the above condition, it is beneficial to controlling temperature drift and thus has good temperature performance.
- (38) The focal length of the fifth lens **L5** is defined as f_5 , and a focal length of the sixth lens **L6** is defined as f_6 . An on-axis distance from an image-side surface of the fifth lens **L5** to an object-side surface of the sixth lens **L6** is defined as d_{10} . Following conditions are satisfied: $d_{10} > 0$; and $0.40 \leq |f_5/f_6| \leq 3.00$. A ratio of the focal length f_5 of the lens **L5** to the focal length f_6 of the lens **L6** is specified. Within the range of the above condition, the absolute values of focal length of two lens of glued lenses are close to each other, it is beneficial to smooth transition of light, improving imaging quality, and causing the chromatic aberration to be $|LC| \leq 10 \mu\text{m}$.
- (39) An on-axis distance from the image-side surface of the seventh lens **L7** to the image plane is defined as BF , and an on-axis distance from the image-side surface of the first lens **L1** to the image-side surface of the seventh lens **L7** is TL . A following condition is satisfied: $0.35 \leq BF/TL$. This condition specifies a ratio of the on-axis distance BF from the image-side surface of the seventh lens **L7** to the image plane to the on-axis distance from the image-side surface of the first lens **L1** to the image-side surface of the seventh lens **L7**. On the basis of miniaturization, the back focal length is long, which is beneficial to assembling the modules. A length TL of lens group **TL** is short, the structure is compact, which is beneficial to reducing the sensitivity of the lens to MTF, improving the production yield and reducing the production cost.
- (40) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL , an on-axis distance from an image-side surface of the second lens **L2** to an aperture **S1** is defined as d_4 , and an on-axis distance from the aperture **S1** to an object-side surface of the third lens **L3** is defined d_0 . A following condition is satisfied: $0.01 \leq (d_4 + d_0)/TTL \leq 0.10$. This condition specifies a ratio of the on-axis distance from the second lens **L2** to the third lens **L3** to the total length TTL of the camera optical lens. Within the range of the above condition, the distance between two lenses adjacent to the aperture **S1** is increased, which is beneficial to improving the imaging quality.
- (41) The field of view of the camera optical lens **10** is defined as FOV , and the image height of the camera optical lens **10** is defined as IH . A light passing diameter of an object-side surface of the first lens **L1** corresponding to a maximum field of view of the camera optical lens **10** is defined as D . A following condition is satisfied: $0 < D/IH/FOV \leq 0.10$. Within the range of the above condition, it is beneficial to achieving miniaturization and small diameter of front opening.
- (42) The field of view of the camera optical lens **10** is defined as FOV , the image height of the camera optical lens **10** is defined as IH , and the optical total length of the camera optical lens is TTL . A following condition is satisfied: $0 < TTL/IH/FOV \leq 0.06$. Within the range of the above condition, it is beneficial to achieving miniaturization and small diameter of front opening.
- (43) An on-axis distance from an image-side surface of the third lens **L3** to an object-side surface of the fourth lens **L4** is defined as d_6 , and an on-axis distance from an image-side surface of the fifth lens **L5** to an object-side surface of the sixth lens **L6** is defined as d_{10} . A following condition is satisfied: $d_6 = 0$; $d_{10} = 0$. The use of two sets of glued lenses is not only beneficial to correcting aberration to achieve high resolution, but also to reducing the problems of tolerance sensitivity of lens units such as tilt/core shift caused in the assembling process.
- (44) In this embodiment, the first lens **L1** has a negative refractive power. An object-side surface of the first lens **L1** is convex in a paraxial region, and an image-side surface of the first lens **L1** is concave in the paraxial region. In other embodiments, the first lens **L1** may also have a positive refractive power, and the object-side surface and image-side surface of the first lens **L1** may also have other concave or convex distribution in the paraxial region.
- (45) The focal length of the camera optical lens **10** is defined as f , and the focal length of the first

lens L1 is defined as f_1 . A following condition is satisfied: $-3.09 \leq f_1/f \leq -0.60$. This condition specifies a ratio of the focal length f_1 of the first lens L1 to the focal length f of the camera optical lens 10. Within the range of the above condition, the first lens has an appropriate negative refractive power, which is beneficial to reducing the aberration of the system while developing ultra-thinness and wide angle of the lens. Optionally, a following condition is satisfied: $-1.93 \leq f_1/f \leq -0.75$.

(46) A central curvature radius of the object-side surface of the first lens L1 is defined as R_1 , and a central curvature radius of the image-side surface of the first lens L1 is R_2 . A following condition is satisfied: $0.57 \leq (R_1 + R_2)/(R_1 - R_2) \leq 1.88$. The shape of the first lens L1 is controlled reasonably, so that the first lens L1 can effectively correct the spherical aberration of the camera optical lens 10. Optionally, a following condition is satisfied: $0.91 \leq (R_1 + R_2)/(R_1 - R_2) \leq 1.51$.

(47) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens 10 along an optic axis is defined as TTL, and an on-axis thickness of the first lens L1 is defined as d_1 . A following condition is satisfied: $0.02 \leq d_1/\text{TTL} \leq 0.05$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.03 \leq d_1/\text{TTL} \leq 0.04$.

(48) In this embodiment, the second lens L2 has a positive refractive power. An object-side surface of the second lens L2 is concave in a paraxial region, and an image-side surface of the second lens L2 is convex in the paraxial region. In other embodiments, the second lens L2 may also have a negative refractive power, and the object-side surface and image-side surface of the second lens L2 may also have other concave or convex distribution in the paraxial region.

(49) The focal length of the camera optical lens 10 is defined as f , and the focal length of the second lens L2 is defined as f_2 . A following condition is satisfied: $-10.46 \leq f_2/f \leq 19.43$. By controlling the optical focal length of the second lens L2 within a reasonable range, it is beneficial to correct the aberration of the camera optical lens 10. Optionally, a following condition is satisfied: $-6.54 \leq f_2/f \leq 15.54$.

(50) A central curvature radius of the object-side surface of the second lens L2 is defined as R_3 , and a central curvature radius of the image-side surface of the second lens L2 is defined as R_4 . A following condition is satisfied: $-4.44 \leq (R_3 + R_4)/(R_3 - R_4) \leq 22.86$, which specifies the shape of the second lens L2. Within the range of the above condition, with the development of the camera optical lens 10 to ultra-thinness and wide angle, it is beneficial to correcting the on-axis chromatic aberration. Optionally, a following condition is satisfied:

(51) $-2.78 \leq (R_3 + R_4)/(R_3 - R_4) \leq 18.29$.

(52) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens 10 along an optic axis is defined as TTL, and the on-axis thickness of the second lens L2 is defined as d_3 . A following condition is satisfied: $0.03 \leq d_3/\text{TTL} \leq 0.09$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.04 \leq d_3/\text{TTL} \leq 0.07$.

(53) In this embodiment, the third lens L3 has a negative refractive power. An object-side surface of the third lens L3 is concave in a paraxial region, and an image-side surface of the third lens L3 is concave in the paraxial region. In other embodiments, the third lens L3 may also have a positive refractive power, and the object-side surface and image-side surface of the third lens L3 may also have other concave or convex distribution in the paraxial region.

(54) The focal length of the camera optical lens 10 is defined as f , and the focal length of the third lens L3 is defined as f_3 . A following condition is satisfied: $-9.86 \leq f_3/f \leq 1.77$. The optical focal length is controlled reasonably, so that the camera optical lens 10 has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied:

(55) $-6.17 \leq f_3/f \leq 1.42$.

(56) A central curvature radius of the object-side surface of the third lens L3 is defined as R_5 , and a central curvature radius of the image-side surface of the third lens L3 is defined as R_6 . A following

condition is satisfied: $-1.65 \leq (R5+R6)/(R5-R6) \leq 0.53$, which specifies the shape of the third lens L3. Within the range of the above condition, it is beneficial to molding the third lens L3, which can alleviate the deflection degree of light passing through the lens and effectively reduce the aberration. Optionally, a following condition is satisfied:

(57) $-1.03 \leq (R5 + R6) / (R5 - R6) \leq 0.43$.

(58) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens 10 along an optic axis is defined as TTL, and an on-axis thickness of the third lens L3 is defined as d5. A following condition is satisfied: $0.01 \leq d5/TTL \leq 0.14$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d5/TTL \leq 0.11$.

(59) In this embodiment, the fourth lens L4 has a positive refractive power. The object-side surface of the fourth lens L4 is convex in the paraxial region, and the image side of the fourth lens L4 is convex in the paraxial region. In other embodiments, the fourth lens L4 may also have a negative refractive power, and the object-side surface and image-side surface of the fourth lens L4 may also have other concave or convex distribution in the paraxial region.

(60) The focal length of the camera optical lens 10 is defined as f, and a focal length of the fourth lens L4 is defined as f4. A following condition is satisfied: $-3.27 \leq f4/f \leq 2.38$, which specifies a ratio of the focal length f4 of the fourth lens L4 to the focal length f of the camera optical lens 10. The optical focal length is controlled reasonably, so that the camera optical lens 10 has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied: $-2.05 \leq f4/f \leq 1.90$.

(61) A central curvature radius of the object-side surface of the fourth lens L4 is defined as R7, and a central curvature radius of the image-side surface of the fourth lens L4 is defined as R8. A following condition is satisfied: $-0.14 \leq (R7+R8)/(R7-R8) \leq 1.27$, which specifies the shape of L4 of the fourth lens. Within the range of the above condition, with the development of ultra-thinness and wide angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-0.09 \leq (R7+R8)/(R7-R8) \leq 1.01$.

(62) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens 10 along an optic axis is defined as TTL, and an on-axis thickness of the fourth lens L4 is defined as d7. A following condition is satisfied: $0.01 \leq d7/TTL \leq 0.23$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d7/TTL \leq 0.19$.

(63) In this embodiment, the fifth lens L5 has a positive refractive power. The object-side surface of the fifth lens L5 is convex in the paraxial region, and the image side of the fifth lens L5 is convex in the paraxial region. In other embodiments, the fifth lens L5 may also have a negative refractive power, and the object-side surface and image-side surface of the fifth lens L5 may also have other concave or convex distribution in the paraxial region.

(64) A central curvature radius of the object-side surface of the fifth lens L5 is defined as R9, and a central curvature radius of the image-side surface of the fifth lens L5 is defined as R10. A following condition is satisfied: $-0.42 \leq (R9+R10)/(R9-R10) \leq 0.84$, which specifies the shape of the fifth lens L5. Within the range of the above condition, with the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-0.26 \leq (R9+R10)/(R9-R10) \leq 0.67$.

(65) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens 10 along an optic axis is defined as TTL, and an on-axis thickness of the fifth lens L5 is defined as d9. A following condition is satisfied: $0.05 \leq d9/TTL \leq 0.22$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.09 \leq d9/TTL \leq 0.17$.

(66) In this embodiment, the sixth lens L6 has a negative refractive power. The object-side surface of the sixth lens L6 is concave in the paraxial region, and the image side of the sixth lens L6 is concave in the paraxial region. In other embodiments, the sixth lens L6 may also have a positive

refractive power, and the object-side surface and image-side surface of the sixth lens L6 may also have other concave or convex distribution in the paraxial region.

(67) The focal length of the optical lens **10** is defined as f , and the focal length of the sixth lens L6 is defined as f_6 . A following condition is satisfied: $-2.88 \leq f_6/f \leq -0.76$. The optical focal length is controlled reasonably, so that the camera optical lens **10** has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied: $-1.80 \leq f_6/f \leq -0.95$.

(68) A central curvature radius of the object-side surface of the sixth lens L6 is defined as R_{11} and a central curvature radius of the image-side surface of the sixth lens L6 is defined as R_{12} . A following condition is satisfied: $-2.78 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 0.42$, which specifies the shape of the sixth lens L6. Within the range of the above condition, with the development of ultra-thinness and wide angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-1.74 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 0.33$.

(69) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the sixth lens L6 is defined as d_{11} . A following condition is satisfied: $0.01 \leq d_{11}/TTL \leq 0.16$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d_{11}/TTL \leq 0.12$.

(70) In this embodiment, the seventh lens L7 has a positive refractive power. The object-side surface of the seventh lens L7 is convex in the paraxial region, and the image side of the seventh lens L7 is convex in the paraxial region. In other embodiments, the seventh lens L7 may also have a negative refractive power, and the object-side surface and image-side surface of the seventh lens L7 may also have other concave or convex distribution in the paraxial region.

(71) A central curvature radius of the object-side surface of the seventh lens L7 is defined as R_{13} , and a central curvature radius of the image-side surface of the seventh lens L7 is defined as R_{14} . A following condition is satisfied: $-0.50 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 0.28$, which specifies the shape of the seventh lens L7. Within the range of the above condition with the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied:

(72) $-0.31 \leq (R_{13} + R_{14}) / (R_{13} - R_{14}) \leq 0.23$.

(73) A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the seventh lens L7 is defined as d_{13} . A following condition is satisfied: $0.05 \leq d_{13}/TTL \leq 0.20$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.08 \leq d_{13}/TTL \leq 0.16$.

(74) In this embodiment, the F number of the camera optical lens **10** is defined as FNO. A following condition is satisfied: $FNO \leq 1.75$, so that it is beneficial to achieving a large aperture. Optionally, a following condition is satisfied: $FNO \leq 1.72$.

(75) In this embodiment, the F number of the camera optical lens **10** is defined as FNO, and a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL. A following condition is satisfied: $TTL/FNO \leq 5.00$, achieving miniaturization and small diameter of front opening.

(76) When the above conditions are satisfied, the camera optical lens **10** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost. According to properties of the camera optical lens **10**, the camera optical lens **10** is especially suitable for mobile phone camera lens components and WEB camera lens that are composed of high-pixel CCD, CMOS and other imaging elements, respectively.

(77) The camera optical lens **10** of the present disclosure will be described below with examples. The symbols recorded in each example will be described as follows. The focal length, on-axis

distance, central curvature radius, on-axis thickness, inflection point, and arrest point are each in unit of millimeter (mm).

(78) TTL denotes a total optical length (on-axis distance from the object-side surface of the first lens L1 to the image plane Si), with a unit of millimeter (mm).

(79) F number FNO denotes a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.

(80) Optionally, the object-side surface and/or the image-side surface of the lens may be provided with inflection points and/or arrest points in order to meet high-quality imaging requirements. The description below may be referred to in specific embodiments as follows.

(81) Design data of the camera optical lens 10 according to Embodiment 1 of the present disclosure are shown in Tables 1 and 2.

(82) TABLE-US-00001 TABLE 1 R d nd vd R1 58.524 d1= 1.000 nd1 1.7684 v1 49.29 R2 6.614 d2= 1.998 R3 -13.439 d3= 1.611 nd2 1.8100 v2 41.00 R4 -11.783 d4= 0.500 S1 ∞ d0= 0.661 R5 -7.342 d5= 0.806 nd3 1.6204 v3 60.37 R6 16.942 d6= 0.000 R7 16.942 d7= 4.500 nd4 1.9537 v4 32.32 R8 -9.598 d8= 0.134 R9 17.864 d9= 3.288 nd5 1.6921 v5 54.57 R10 -12.991 d10= 0.000 R11 -12.991 d11= 0.806 nd6 1.9229 v6 18.90 R12 19.451 d12= 0.600 R13 12.362 d13= 4.000 nd7 1.5891 v7 61.16 R14 -10.000 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 9.297

(83) The reference signs are explained as follows. S1: aperture; R: central curvature radius of an optical surface; R1: central curvature radius of the object-side surface of the first lens L1; R2: central curvature radius of the image-side surface of the first lens L1; R3: central curvature radius of the object-side surface of the second lens L2; R4: central curvature radius of the image-side surface of the second lens L2; R5: central curvature radius of the object-side surface of the third lens L3; R6: central curvature radius of the image-side surface of the third lens L3; R7: central curvature radius of the object-side surface of the fourth lens L4; R8: central curvature radius of the image-side surface of the fourth lens L4; R9: central curvature radius of the object-side surface of the fifth lens L5; R10: central curvature radius of the image-side surface of the fifth lens L5; R11: central curvature radius of the object-side surface of the sixth lens L6; R12: central curvature radius of the image-side surface of the sixth lens L6; R13: central curvature radius of the object-side surface of the seventh lens L7; R14: central curvature radius of the image-side surface of the seventh lens L7; R15: central curvature radius of the object-side surface of the optical filter GF; R16: central curvature radius of the image-side surface of the optical filter GF; d: on-axis thickness of a lens and an on-axis distance between lenses; d0: on-axis distance from the aperture S1 to the object-side surface of the third lens L3; d1: on-axis thickness of the first lens L1; d2: on-axis distance from the image-side surface of the first lens L1 to the object-side surface of the second lens L2; d3: on-axis thickness of the second lens L2; d4: on-axis distance from the image-side surface of the second lens L2 to the aperture S1; d5: on-axis thickness of the third lens L3; d6: on-axis distance from the image-side surface of the third lens L3 to the object-side surface of the fourth lens L4; d7: on-axis thickness of the fourth lens L4; d8: on-axis distance from the image-side surface of the fourth lens L4 to the object-side surface of the fifth lens L5; d9: on-axis thickness of the fifth lens L5; d10: on-axis distance from the image-side surface of the fifth lens L5 to the object-side surface of the sixth lens L6; d11: on-axis thickness of the sixth lens L6; d12: on-axis distance from the image-side surface of the sixth lens L6 to the object-side surface of the seventh lens L7; d13: on-axis thickness of the seventh lens L7; d14: on-axis distance from the image-side surface of the seventh lens L7 to the object-side surface of the optical filter GF; d15: on-axis thickness of the optical filter GF; d16: on-axis distance from the image-side surface of the optical filter GF to the image plane Si; nd: refractive index of a d-line (the d-line is green light with a wavelength of 555 nm); nd1: refractive index of the d-line of the first lens L1; nd2: refractive index of the d-line of the second lens L2; nd3: refractive index of the d-line of the third lens L3; nd4: refractive index of the d-line of the fourth lens L4; nd5: refractive index of the d-line of the

fifth lens L5; nd6: refractive index of the d-line of the sixth lens L6; nd7: refractive index of the d-line of the seventh lens L7; ndg: refractive index of the d-line of the optical filter GF; vd: Abbe number; v1: Abbe number of the first lens L1; v2: Abbe number of the second lens L2; v3: Abbe number of the third lens L3; v4: Abbe number of the fourth lens L4; v5: Abbe number of the fifth lens L5; v6: Abbe number of the sixth lens L6; v7: Abbe number of the seventh lens L7; vg: Abbe number of the optical filter GF.

(84) Table 2 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 10 according to Embodiment 1 of the present disclosure.

(85) It should be noted that that each lens surface of the third lens L3, the fourth lens L4, the fifth lens L5 and the sixth lens L6 in the camera optical lens 10 in Embodiment 1 is a spherical surface.

(86) TABLE-US-00002 TABLE 2 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12 R1

1.2702E+02	1.9741E-03	-1.3814E-04	6.1484E-06	-1.4288E-07	9.0215E-10	R2
5.4438E-01	2.9063E-03	-6.3457E-05	-6.4599E-06	1.2949E-06	-2.3331E-08	R3
3.3777E+00	-1.5381E-04	4.2920E-06	-4.6369E-06	9.6074E-07	-4.2277E-08	R4
-1.2799E-01	-8.3240E-05	9.2990E-06	-3.2090E-06	6.4871E-07	-3.2714E-08	R13
-2.4460E+00	-5.1900E-06	1.9260E-07	3.5846E-08	-2.7044E-10	6.0425E-12	R14
-2.1602E+00	1.3547E-05	9.4447E-09	4.9161E-08	-5.3020E-10	1.3417E-11	

(87) For convenience, the aspheric surfaces of the first lens L1, the second lens L2, and the seventh lens L7 use the aspheric surfaces shown in formula (1) below. However, the present disclosure is not limited to the polynomial form of aspheric surfaces represented by formula (1).

$$(88) \quad z = (cr^2) / \{1 + [1(k + 1)(c^2 r^2)]^{1/2}\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} \quad (1)$$

(89) Here, k denotes a conic coefficient, and A4, A6, A8, A10, and A12 each denote an aspheric coefficient, respectively, c denotes an curvature at a center of an optical surface, r denotes a vertical distance between a point on a curve of the aspheric surface and the optical axis, and Z denotes an aspheric depth (i.e., a vertical distance between a point of the aspheric surface from which a distance to the optical axis is r, and a tangent plane tangent to a vertex on an aspheric optical axis).

(90) Design data of the inflection point and the arrest point of each lens in the camera optical lens 10 according to Embodiment 1 of the present disclosure are shown in Tables 3 and 4. Here, P1R1 and P1R2 denote the object side surface and image side surface of the first lens L1, respectively. P2R1 and P2R2 denote the object side surface and image side surface of the second lens L2, respectively. P3R1 and P3R2 denote the object side surface and image side surface of the third lens L3, respectively. P4R1 and P4R2 denote the object side surface and image side surface of the fourth lens L4, respectively. P5R1 and P5R2 denote the object side surface and image side surface of the fifth lens L5, respectively. P6R1 and P6R2 denote the object side surface and image side surface of the sixth lens L6, respectively. P7R1 and P7R2 denote the object side surface and image side surface of the seventh lens L7, respectively. Data in an “inflection point” column are a vertical distance from an inflexion point provided on a surface of each lens to the optical axis of the camera optical lens 10. Data in an “arrest point” column are a vertical distance from an arrest point provided on the surface of each lens to the optical axis of the camera optical lens 10.

(91) TABLE-US-00003 TABLE 3 Number of inflection points Inflection point 1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 1 5.285

(92) TABLE-US-00004 TABLE 4 Number of arrest points Arrest point1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 0 /

(93) FIG. 2 and FIG. 3 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 10 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 10 according to Embodiment 1, respectively. FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens 10 after light

having a wavelength of 555 nm passes through the camera optical lens **10** according to Embodiment 1. A field curvature S in FIG. **4** is a field curvature in a sagittal direction, and T is a field curvature in a meridian direction.

(94) Table 29 below shows numerical values corresponding to various numerical values in Embodiments 1, 2, 3, 4 and 5, and parameters specified in the conditions. As shown in Table 29, Embodiment 1 satisfies various conditions.

(95) In this embodiment, the camera optical lens **10** has an entrance pupil diameter ENPD of 3.950 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 102.40° in a diagonal direction. The camera optical lens **10** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 2

(96) FIG. **5** is a structural schematic diagram of a camera optical lens **20** according to Embodiment 2 of the present disclosure. Embodiment 2 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(97) In this embodiment, the object-side surface of the second lens L2 is convex in the paraxial region, and the image-side surface of the second lens L2 is concave in the paraxial region.

(98) Table 5 and Table 6 show the design data of the camera optical lens **20** of the Embodiment 2 of the present disclosure.

(99) TABLE-US-00005 TABLE 5 R d nd vd R1 50.000 d1= 1.000 nd1 1.7725 v1 49.61 R2 4.953 d2= 2.000 R3 17.259 d3= 1.611 nd2 1.7433 v2 49.40 R4 50.000 d4= 1.000 S1 ∞ d0= 1.000 R5 -7.519 d5= 0.806 nd3 1.5174 v3 52.19 R6 78.254 d6= 0.000 R7 78.254 d7= 2.551 nd4 1.9537 v4 32.32 R8 -8.504 d8= 0.133 R9 14.042 d9= 3.974 nd5 1.6921 v5 54.57 R10 -21.404 d10= 0.000 R11 -21.404 d11= 0.806 nd6 1.9229 v6 18.90 R12 12.093 d12= 0.300 R13 10.516 d13= 3.370 nd7 1.6180 v7 63.41 R14 -9.803 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 8.826

(100) Table 6 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens **20** according to Embodiment 2 of the present disclosure.

(101) TABLE-US-00006 TABLE 6 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12 R1 7.3818E+01 1.5470E-04 2.3928E-06 -9.0391E-08 -6.0759E-09 -5.6886E-11 R2 2.7310E-01 3.5301E-04 4.9216E-05 1.1979E-06 1.5973E-07 4.0319E-08 R3 -4.1838E+00 1.3787E-04 2.0759E-05 3.0865E-06 2.4227E-07 6.6851E-09 R4 6.4431E+01 4.9935E-05 1.3584E-05 1.3183E-06 1.2994E-07 1.0727E-08 R13 -2.1026E+00 -2.8727E-06 4.9172E-07 2.7468E-08 1.3838E-09 3.1063E-11 R14 -1.9136E+00 -9.6042E-06 -1.5748E-07 8.6307E-09 9.6703E-10 9.0303E-11

(102) Design data of the inflection point and the arrest point of each lens in the camera optical lens **20** according to Embodiment 2 of the present disclosure are shown in Tables 7 and 8.

(103) TABLE-US-00007 TABLE 7 Number of inflection points Inflection point 1 P1R1 1 4.425 P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 1 4.665

(104) TABLE-US-00008 TABLE 8 Number of arrest points Arrest point1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 0 /

(105) FIG. **6** and FIG. **7** are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens **20** after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens **20** according to Embodiment 2, respectively. FIG. **8** is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens **20** according to Embodiment 2. The field curvature S in

FIG. 8 is a field curvature in a sagittal direction, and Tis a field curvature in a meridional direction. (106) As shown in Table 29, Embodiment 2 satisfies various conditions. (107) In this embodiment, the camera optical lens 20 has an entrance pupil diameter ENPD of 3.930 mm, a full-field image height IH of 5.145 mm, and a field of view FOV 102.10° in a diagonal direction. The camera optical lens 20 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 3

(108) FIG. 9 is a structural schematic diagram of a camera optical lens 30 according to Embodiment 3 of the present disclosure. Embodiment 3 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(109) In this embodiment, the object-side surface of the third lens L3 is convex in the paraxial region, and the image-side surface of the third lens L3 is convex in the paraxial region. The object-side surface of the fourth lens L4 is concave in the paraxial region, and the image-side surface of the fourth lens L4 is concave in the paraxial region. The image-side surface of the sixth lens L6 is convex in the paraxial region. The third lens L3 has a positive refractive power, and the fourth lens L4 has a negative refractive power.

(110) Design data of the camera optical lens 30 of Embodiment 3 of the present disclosure are shown in Tables 9 and 10.

(111) TABLE-US-00009 TABLE 9 R d nd vd R1 50.000 d1= 1.000 nd1 1.7740 v1 49.60 R2 4.456 d2= 2.940 R3 -30.000 d3= 1.611 nd2 1.8100 v2 41.00 R4 -21.000 d4= 0.600 S1 ∞ d0= 0.000 R5 21.146 d5= 2.551 nd3 1.9537 v3 32.32 R6 -10.070 d6= 0.000 R7 -10.070 d7= 0.800 nd4 1.5174 v4 52.19 R8 11.606 d8= 0.633 R9 10.107 d9= 3.974 nd5 1.6921 v5 54.57 R10 -6.878 d10= 0.000 R11 -6.878 d11= 0.806 nd6 1.9229 v6 18.90 R12 -77.195 d12= 0.205 R13 14.278 d13= 3.359 nd7 1.7433 v7 49.34 R14 -23.865 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 8.248

(112) Table 10 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 30 of the Embodiment 3 of the present disclosure.

(113) TABLE-US-00010 TABLE 10 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12 R1 1.0563E+02 -3.6645E-04 -3.2648E-05 9.9226E-07 2.0032E-08 -1.3970E-09 R2 5.7717E-02 6.9303E-04 -5.4109E-05 -7.7723E-08 1.2845E-07 1.3919E-09 R3 -8.3399E+01 7.2850E-04 -9.8556E-06 1.2989E-06 7.2341E-08 1.9118E-09 R4 -1.4483E+01 1.5478E-04 -1.0031E-05 -8.0541E-07 1.1500E-07 -2.0905E-10 R13 3.8745E+00 2.3772E-04 2.6665E-05 -1.6177E-06 5.9282E-08 -7.5860E-10 R14 -1.1735E+01 9.9744E-04 1.0816E-05 1.7072E-06 -8.8881E-08 2.6969E-09

(114) Design data of the inflection point and the arrest point of each lens in the camera optical lens 30 according to Embodiment 3 of the present disclosure are shown in Tables 11 and 12.

(115) TABLE-US-00011 TABLE 11 Number of inflection points Inflection point 1 Inflection point 2 P1R1 2 1.905 4.455 P1R2 0 // P2R1 1 1.665 / P2R2 1 3.075 / P3R1 0 // P3R2 0 // P4R1 0 // P4R2 0 // P5R1 0 // P5R2 0 // P6R1 0 // P6R2 0 // P7R1 0 // P7R2 1 1.695 /

(116) TABLE-US-00012 TABLE 12 Number of arrest points Arrest point1 P1R1 1 3.195 P1R2 0 / P2R1 1 2.735 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 1 2.795

(117) FIG. 10 and FIG. 11 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 30 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 30 according to Embodiment 3, respectively. FIG. 12 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 30 according to Embodiment 3. The field curvature

S in FIG. 12 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

(118) Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens 30 of this embodiment satisfies the above conditions.

(119) In this embodiment, the camera optical lens 30 has an entrance pupil diameter ENPD of 3.915 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 102.60° in a diagonal direction. The camera optical lens 30 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 4

(120) FIG. 13 is a structural schematic diagram of a camera optical lens 40 according to Embodiment 4 of the present disclosure. Embodiment 4 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(121) In this embodiment, the object-side surface of the second lens L3 is convex in the paraxial region, and the image-side surface of the sixth lens L6 is convex in the paraxial region.

(122) Design data of the camera optical lens 40 of Embodiment 4 of the present disclosure are shown in Tables 13 and 14.

(123) TABLE-US-00013 TABLE 13 R d nd vd R1 68.688 d1= 1.000 nd1 1.8839 v1 37.21 R2 4.605 d2= 3.426 R3 50.000 d3= 1.700 nd2 1.8839 v2 37.21 R4 -24.531 d4= 0.100 S1 ∞ d0= 0.367 R5 -22.927 d5= 0.806 nd3 1.6204 v3 60.37 R6 120.000 d6= 0.000 R7 120.000 d7= 4.500 nd4 1.9537 v4 32.32 R8 -10.132 d8= 0.134 R9 11.358 d9= 4.000 nd5 1.4970 v5 81.59 R10 -6.931 d10= 0.000 R11 -6.931 d11= 0.806 nd6 1.9229 v6 18.90 R12 -42.255 d12= 0.400 R13 28.200 d13= 3.000 nd7 1.5891 v7 61.16 R14 -29.094 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 7.974

(124) Table 14 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 40 according to Embodiment 4 of the present disclosure.

(125) TABLE-US-00014 TABLE 14 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12
R1 1.2702E+02 2.2951E-04 -9.6203E-05 6.0729E-06 -1.8605E-07 2.2807E-09 R2
1.8838E-01 1.0818E-03 -1.3415E-04 -8.9339E-07 5.9204E-07 -3.1622E-08 R3 1.5889E+02
3.3344E-04 -2.4609E-05 -8.4717E-07 1.3422E-07 -5.8371E-09 R4 -2.2815E+00
1.2354E-04 -1.1096E-05 -6.8666E-07 6.5065E-08 -7.6475E-10 R13 2.4141E+01 9.4874E-04
8.4097E-06 -2.2454E-07 8.9037E-09 -4.4577E-11 R14 -1.3706E+02 8.2668E-04
4.3843E-05 4.9173E-07 -7.7032E-08 3.5685E-09

(126) Design data of the inflection point and the arrest point of each lens in the camera optical lens 40 according to Embodiment 4 of the present disclosure are shown in Tables 15 and 16.

(127) TABLE-US-00015 TABLE 15 Number of inflection points Inflection point 1 P1R1 1 2.005
P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2
0 / P7R1 0 / P7R2 1 1.375

(128) TABLE-US-00016 TABLE 16 Number of arrest points Arrest point1 P1R1 1 3.265 P1R2 0 /
P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1
0 / P7R2 1 2.305

(129) FIG. 14 and FIG. 15 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 40 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 40 according to Embodiment 4, respectively. FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens 40 after light having a wavelength of 555 nm passes through the camera optical lens 40 according to Embodiment 4. A field curvature S in FIG. 16 is a field curvature in a sagittal direction, and T is a

field curvature in a meridian direction.

(130) Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens **40** of this embodiment satisfies the above conditions.

(131) In this embodiment, the camera optical lens **40** has an entrance pupil diameter ENPD of 3.911 mm, a full-field image height IH of 5.145 mm, and a field of view FOV 102.20° in a diagonal direction. The camera optical lens **40** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 5

(132) FIG. **17** is a structural schematic diagram of a camera optical lens **50** according to Embodiment 5 of the present disclosure. Embodiment 5 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(133) In this embodiment, the second lens **L2** has a negative refractive power.

(134) Table 17 and Table 18 show the design data of the camera optical lens **50** of the Embodiment 5 of the present disclosure.

(135) TABLE-US-00017 TABLE 17 R d nd vd R1 58.524 d1= 1.000 nd1 1.8100 v1 41.00 R2 6.394 d2= 2.739 R3 -26.960 d3= 1.611 nd2 1.8100 v2 41.00 R4 -10.562 d4= -0.200 S1 ∞ d0= 0.704 R5 -5.787 d5= 0.806 nd3 1.6204 v3 60.37 R6 14.553 d6= 0.000 R7 14.553 d7= 4.500 nd4 1.9537 v4 32.32 R8 -11.272 d8= 0.134 R9 14.627 d9= 3.288 nd5 1.6921 v5 54.57 R10 -11.087 d10= 0.000 R11 -11.087 d11= 0.806 nd6 1.9229 v6 18.90 R12 26.794 d12= 0.600 R13 14.641 d13= 4.000 nd7 1.5891 v7 61.16 R14 -10.000 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 9.209

(136) Table 18 shows aspheric surface data of the first lens **L1**, the second lens **L2** and the seventh lens **L7** in the camera optical lens **50** according to Embodiment 5 of the present disclosure.

(137) TABLE-US-00018 TABLE 18 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12 R1 1.2702E+02 1.5541E-03 -8.3040E-05 4.1280E-06 -1.0898E-07 8.3786E-10 R2 2.3390E+00 1.5610E-03 -9.4194E-05 9.4331E-06 -7.9449E-07 5.7644E-08 R3 5.2620E+01 -4.8282E-04 1.7696E-04 -2.1534E-05 2.4182E-06 -4.3223E-08 R4 7.3128E-01 -8.7169E-04 1.0347E-04 -2.2897E-05 2.7479E-06 -9.6706E-08 R13 -1.0927E+01 -4.7085E-05 -9.1800E-06 8.8604E-08 1.2889E-09 -1.2247E-10 R14 -2.7369E+00 -1.8288E-04 -3.6583E-06 4.0908E-08 -4.2052E-10 -4.6208E-11

(138) Design data of the inflection point and the arrest point of each lens in the camera optical lens **50** according to Embodiment 5 of the present disclosure are shown in Tables 19 and 20.

(139) TABLE-US-00019 TABLE 19 Number of inflection points Inflection point 1 P1R1 0 / P1R2 0 / P2R1 1 2.255 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 1 3.385 P7R2 0 /

(140) TABLE-US-00020 TABLE 20 Number of arrest points Arrest point1 P1R1 0 / P1R2 0 / P2R1 1 2.945 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 1 5.195 P7R2 0 /

(141) FIG. **18** and FIG. **19** are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens **50** after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens **50** according to Embodiment 5, respectively. FIG. **20** is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens **50** according to Embodiment 5. The field curvature S in FIG. **20** is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

(142) Table 29 below lists the corresponding values of each condition in this embodiment

according to the above conditions. It can be appreciated that the camera optical lens **50** of this embodiment satisfies the above conditions.

(143) In this embodiment, the camera optical lens **50** has an entrance pupil diameter ENPD of 3.943 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 100.80° in a diagonal direction. The camera optical lens **50** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 6

(144) FIG. **21** is a structural schematic diagram of a camera optical lens **30** according to Embodiment 6 of the present disclosure. Embodiment 6 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(145) In this embodiment, the object-side surface of the second lens **L2** is convex in the paraxial region, and the image-side surface of the second lens **L2** is concave in the paraxial region.

(146) Design data of the camera optical lens **60** of Embodiment 6 of the present disclosure are shown in Tables 21 and 22.

(147) TABLE-US-00021 TABLE 21 R d nd vd R1 50.000 d1= 1.000 nd1 1.7433 v1 49.34 R2 4.156 d2= 2.000 R3 17.804 d3= 1.611 nd2 1.7433 v2 49.40 R4 46.943 d4= 1.000 S1 ∞ d0= 1.800 R5 -34.352 d5= 0.806 nd3 1.5174 v3 52.19 R6 21.623 d6= 0.000 R7 21.623 d7= 2.551 nd4 1.9537 v4 32.32 R8 -9.900 d8= 0.133 R9 64.025 d9= 3.974 nd5 1.8040 v5 46.57 R10 -18.191 d10= 0.000 R11 -18.191 d11= 0.806 nd6 1.9229 v6 18.90 R12 10.317 d12= 0.300 R13 9.576 d13= 3.370 nd7 1.6180 v7 63.41 R14 -8.807 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 9.530

(148) Table 22 shows aspheric surface data of the first lens **L1**, the second lens **L2** and the seventh lens **L7** in the camera optical lens **60** according to Embodiment 6 of the present disclosure.

(149) TABLE-US-00022 TABLE 22 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12
R1 9.4237E+01 3.8021E-04 -1.6748E-05 -1.4253E-06 7.5532E-08 -1.3466E-09 R2
1.5000E-01 8.2072E-04 6.2833E-05 -2.7321E-06 6.4190E-07 -6.8938E-08 R3 2.3873E+01
1.0533E-03 9.4246E-05 5.6581E-06 -5.1195E-07 2.0080E-08 R4 2.3909E+02 1.1012E-03
1.0540E-04 -1.8156E-06 3.9743E-08 3.5081E-08 R13 -1.6031E+00 -5.8426E-07
5.0444E-06 -2.0656E-07 -1.6793E-09 6.3518E-10 R14 -2.4235E+00 -1.3177E-04
-6.9161E-06 1.0693E-06 -6.8456E-08 1.9823E-09

(150) Design data of the inflection point and the arrest point of each lens in the camera optical lens **60** according to Embodiment 6 of the present disclosure are shown in Tables 23 and 24.

(151) TABLE-US-00023 TABLE 23 Number of inflection points Inflection point 1 Inflection point 2 P1R1 2 3.305 4.715 P1R2 0 // P2R1 0 // P2R2 0 // P3R1 0 // P3R2 0 // P4R1 0 // P4R2 0 // P5R1 0 // P5R2 0 // P6R1 0 // P6R2 0 // P7R1 0 // P7R2 0 //

(152) TABLE-US-00024 TABLE 24 Number of arrest points Arrest point1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 0 /

(153) FIG. **22** and FIG. **23** are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens **60** after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens **60** according to Embodiment 6, respectively. FIG. **24** is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens **60** according to Embodiment 6. The field curvature S in FIG. **24** is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

(154) Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens **60** of this

embodiment satisfies the above conditions.

(155) In this embodiment, the camera optical lens **60** has an entrance pupil diameter ENPD of 3.586 mm, a full-field image height IH of 5.145 mm, and a field of view FOV 102.20° in a diagonal direction. The camera optical lens **60** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Comparative Example 1

(156) FIG. **25** is a structural schematic diagram of a camera optical lens **70** according to Comparative Example 1 of the present disclosure. Comparative Example 1 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

(157) In Comparative Example 1, the second lens **L2** has a negative refractive power. The object-side surface of the second lens **L2** is convex in the paraxial region, and the image-side surface of the second lens **L2** is concave in the paraxial region.

(158) Design data of the camera optical lens **70** of Comparative Example 1 of the present disclosure are shown in Tables 25 and 26.

(159) TABLE-US-00025
TABLE 25 R d nd vd R1 50.000 d1= 1.000 nd1 1.7725 v1 49.61 R2 5.323 d2= 2.000 R3 16.938 d3= 1.611 nd2 1.7433 v2 49.40 R4 49.397 d4= 1.000 S1 ∞ d0= 1.000 R5 -6.112 d5= 0.806 nd3 1.5174 v3 52.19 R6 18.922 d6= 0.000 R7 18.922 d7= 2.551 nd4 1.9537 v4 32.32 R8 -9.689 d8= 0.133 R9 15.061 d9= 3.974 nd5 1.6921 v5 54.57 R10 -18.419 d10= 0.000 R11 -18.419 d11= 0.806 nd6 1.9229 v6 18.90 R12 11.079 d12= 0.300 R13 8.745 d13= 3.370 nd7 1.6180 v7 63.41 R14 -7.833 d14= 0.403 R15 ∞ d15= 0.400 ndg 1.5168 vg 64.21 R16 ∞ d16= 8.580

(160) Table 26 shows aspheric surface data of the first lens **L1**, the second lens **L2** and the seventh lens **L7** in the camera optical lens **70** of the Comparative Example 1 of the present disclosure.

(161) TABLE-US-00026
TABLE 26 Conic coefficient Aspheric coefficient k A4 A6 A8 A10 A12
R1 8.8071E+01 6.7896E-04 -2.1167E-05 3.3667E-08 1.3424E-08 -5.1776E-10 R2 4.7917E-01 4.9019E-04 -2.9033E-06 2.9396E-06 -3.6912E-07 3.0765E-08 R3 -5.9390E+00 1.3867E-04 1.8942E-05 3.0625E-06 2.4364E-07 6.7089E-09 R4 2.4263E+02 2.3132E-05 9.8727E-06 1.7878E-06 1.7838E-07 1.9753E-08 R13 -2.7427E+00 -4.8109E-05 7.0907E-07 -1.8973E-08 2.5524E-10 -6.0620E-11 R14 -1.9943E+00 -5.5232E-05 -6.9793E-06 1.7321E-07 3.5475E-09 -2.4854E-10

(162) Design data of the inflection point and the arrest point of each lens in the camera optical lens **70** according to Comparative Example 1 of the present disclosure are shown in Tables 27 and 78

(163) TABLE-US-00027
TABLE 27 Number of inflection points Inflection point 1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 1 4.805 P7R2 0 /

(164) TABLE-US-00028
TABLE 28 Number of arrest points Arrest point1 P1R1 0 / P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 0 / P6R1 0 / P6R2 0 / P7R1 0 / P7R2 0 /

(165) FIG. **26** and FIG. **27** are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens **70** after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens **70** according to Comparative Example 1, respectively. FIG. **28** is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens **70** according to Comparative Example 1. The field curvature **S** in FIG. **28** is a field curvature in a sagittal direction, and **T** is a field curvature in a meridional direction.

(166) Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens **70** of

Comparative Example 1 does not satisfy the following condition: $1.30 \leq f_7/f \leq 4.00$.

(167) In Comparative Example 1, the camera optical lens **70** has an entrance pupil diameter ENPD of 3.713 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 104.20° in a diagonal direction. The aberration of the camera optical lens **70** is not fully corrected, which is not beneficial to reducing the diameter of front opening, so that its optical performance is not excellent enough.

(168) TABLE-US-00029 TABLE 29 Parameters and Comparative conditions

Embodiment 1	Embodiment 2	Embodiment 3	Embodiment 4	Embodiment 5	Embodiment 6	Example 1	f7/f
1.39	1.99	3.95	1.69	1.31	1.22	n1	1.77
1.77	1.77	1.77	1.88	1.81	1.74	1.77	f34/f
2.89	2.52	3.44	2.08	6.95	3.12	3.33	(FOV × f)/IH
125.79	124.81	124.92	124.31	124.09	121.09	120.31	f
6.320	6.289	6.264	6.258	6.334	6.096	5.940	f1
-9.750	-7.160	-6.358	-5.598	-8.899	-6.133	-7.760	f2
81.851	34.602	79.643	18.730	-33.138	37.558	-9.131	f3
-8.128	-13.169	7.409	-30.866	-6.554	-25.480	-8.801	f4
6.967	8.114	-10.255	9.909	7.241	7.373	6.986	f5
11.326	12.799	6.519	9.319	9.584	17.938	12.542	f6
-8.262	-8.200	-8.150	-8.998	-8.333	-6.972	-7.329	f7
10.020	8.741	12.442	24.715	10.701	7.960	7.229	FNO
1.600	1.600	1.600	1.600	1.606	1.700	1.600	TTL
30.004	28.180	27.530	29.015	30.000	29.684	27.934	IH
5.145	5.145	5.145	5.145	5.145	5.145	5.145	FOV
102.40°	102.10°	102.60°	102.20°	100.80°	102.20°	104.20°	D
4.617	4.763	4.545	4.384	4.884	5.009	4.956	f34
18.254	15.880	21.553	13.000	44.000	18.99	19.756	

(169) FOV denotes a field of view of the camera optical lens in a diagonal direction.

(170) The above are only preferred embodiments of the present disclosure. Here, it should be noted that those skilled in the art may make modifications without departing from the inventive concept of the present disclosure, but these shall fall into the protection scope of the present disclosure.

Claims

1. A camera optical lens, comprising from an object side to an image side: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; and a seventh lens, wherein the camera optical lens satisfies following conditions: $1.3 \leq f_7/f \leq 4$; $1.7 \leq n_1$; $2 \leq f_{34}/f \leq 7$; and $120 \leq (\text{FOV} \times f)/\text{IH}$ where f denotes a focal length of the camera optical lens; f_7 denotes a focal length of the seventh lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; n_1 denotes a refractive index of the first lens; FOV denotes a field of view of the camera optical lens; and IH denotes an image height of the camera optical lens.
2. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition: $1 \leq f_5/f \leq 3$; where f_5 denotes a focal length of the fifth lens.
3. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions: $d_{10} = 0$; and $0.4 \leq f_5/f_6 \leq 3$, where d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens; f_5 denotes a focal length of the fifth lens; and f_6 denotes a focal length of the sixth lens.
4. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition: $0.35 \leq \text{BF}/\text{TL}$, where BF denotes an on-axis distance from an image-side surface of the seventh lens to an image plane of the camera optical lens; and TL denotes an on-axis distance from an object-side surface of the first lens to the image-side surface of the seventh lens.
5. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition: $0.01 \leq (d_4 + d_0)/\text{TTL} \leq 0.1$; where d_4 denotes an on-axis distance from an image-side surface of the second lens to an aperture; d_0 denotes an on-axis distance from the aperture to an object-side surface of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
6. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition: $0 < D/\text{IH}/\text{FOV} \leq 0.1$; where D denotes a diameter of a light passing

opening of an object-side surface of the first lens corresponding to a maximum field of view of the camera optical lens.

7. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition: $0 < \text{TTL} / \text{IH} / \text{FOV} \leq 0.06$; where TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
8. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions: $d_6 = 0$; and $d_{10} = 0$, where d_6 denotes an on-axis distance from an image-side surface of the third lens to an object-side surface of the fourth lens; and d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens.
9. The camera optical lens as described in claim 1, wherein an object-side surface of the first lens is convex in a paraxial region, and an image-side surface of the first lens is concave in the paraxial region; the first lens has a negative refractive power; and the camera optical lens satisfies following conditions: $-3.09 \leq f_1 / f \leq -0.6$; $0.57 \leq (R_1 + R_2) / (R_1 - R_2) \leq 1.88$; and $0.02 \leq d_1 / \text{TTL} \leq 0.05$, where f_1 denotes a focal length of the first lens; R_1 denotes a central curvature radius of the object-side surface of the first lens; R_2 denotes a central curvature radius of the image-side surface of the first lens; d_1 denotes an on-axis thickness of the first lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
10. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions: $-10.46 \leq f_2 / f \leq 19.43$; $-4.44 \leq (R_3 + R_4) / (R_3 - R_4) \leq 22.86$; and $0.03 \leq d_3 / \text{TTL} \leq 0.09$, where f_2 denotes a focal length of the second lens; R_3 denotes a central curvature radius of an object-side surface of the second lens; R_4 denotes a central curvature radius of an image-side surface of the second lens; d_3 denotes an on-axis thickness of the second lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
11. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions: $-9.86 \leq f_3 / f \leq 1.77$; $-1.65 \leq (R_5 + R_6) / (R_5 - R_6) \leq 0.53$; and $0.01 \leq d_5 / \text{TTL} \leq 0.14$, where f_3 denotes a focal length of the third lens; R_5 denotes a central curvature radius of an object-side surface of the third lens; R_6 denotes a central curvature radius of an image-side surface of the third lens; d_5 denotes an on-axis thickness of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
12. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions: $-3.27 \leq f_4 / f \leq 2.38$; $-0.14 \leq (R_7 + R_8) / (R_7 - R_8) \leq 1.27$; and $0.01 \leq d_7 / \text{TTL} \leq 0.23$, where f_4 denotes a focal length of the fourth lens; R_7 denotes a central curvature radius of an object-side surface of the fourth lens; R_8 denotes a central curvature radius of an image-side surface of the fourth lens; d_7 denotes an on-axis thickness of the fourth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
13. The camera optical lens as described in claim 1, wherein an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of the fifth lens is convex in the paraxial region; the fifth lens has positive refractive power; the camera optical lens satisfies following conditions: $-0.42 \leq (R_9 + R_{10}) / (R_9 - R_{10}) \leq 0.84$; and $0.05 \leq d_9 / \text{TTL} \leq 0.22$, where R_9 denotes a central curvature radius of an object-side surface of the fifth lens; R_{10} denotes a central curvature radius of an image-side surface of the fifth lens; d_9 denotes an on-axis thickness of the fifth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.
14. The camera optical lens as described in claim 1, wherein an object-side surface of the sixth lens is concave in a paraxial region; the sixth lens has negative refractive power; the camera optical lens satisfies following conditions: $-2.88 \leq f_6 / f \leq -0.76$;

$-2.78 \leq (R11 + R12) / (R11 - R12) \leq 0.42$; and $0.01 \leq d11 / TTL \leq 0.16$, where $f6$ denotes a focal length of the sixth lens; $R11$ denotes a central curvature radius of an object-side surface of the sixth lens; $R12$ denotes a central curvature radius of an image-side surface of the sixth lens; $d11$ denotes an on-axis thickness of the sixth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

15. The camera optical lens as described in claim 1, wherein an object-side surface of the seventh lens is convex in a paraxial region, and an image-side surface of the seventh lens is convex in the paraxial region; the seventh lens has positive refractive power; the camera optical lens satisfies following conditions: $-0.50 \leq (R13 + R14) / (R13 - R14) \leq 0.28$; and $0.05 \leq d13 / TTL \leq 0.2$, where $R13$ denotes a central curvature radius of an object-side surface of the seventh lens; $R14$ denotes a central curvature radius of an image-side surface of the seventh lens; $d13$ denotes an on-axis thickness of the seventh lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

16. The camera optical lens as described in claim 1, wherein at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, and the seventh lens is made of glass material.
